

PC-2025/2026 Second Assignment

Bloom Filter: Parallel Optimization and Asynchronous Pipelines

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Abstract

This report exposes a project developed in the context of Parallel Computing course. The study presented consists in the implementation of a Bloom Filter and on the study of its optimization using parallel algorithms and asynchronous interactions between components. This project has been developed using Python as primary language and exploiting the possibility offered by this instrument in terms of multiprocessing capability, multithreading and asynchronous features. Specifically, this study introduces a multi-tiered parallelization strategy to tackle the memory bandwidth and concurrency bottlenecks inherent in massive-scale probabilistic data structures. The proposed architectures evolve from traditional GIL-bound multiprocessing—utilizing low-overhead Inter-Process Communication (IPC)—to cutting-edge Free-Threaded (No-GIL) execution paradigms. To maximize hardware utilization and minimize thread contention, advanced memory management techniques, including cache-aligned Striped Bloom Filters and Structure of Arrays (SoA) layouts, were designed and integrated alongside asynchronous I/O pipelines to effectively mask disk latency. The resulting implementations were rigorously benchmarked under both strong (Amdahl's Law) and weak (Gustafson's Law) scaling regimes, utilizing large-scale synthetic workloads and real-world datasets from the Common Crawl corpus. Experimental results demonstrate the high efficiency of the proposed solutions: the asynchronous I/O overlap successfully saturated storage throughput, while the highly optimized No-GIL architectures achieved a peak speedup of 2.5x over the sequential baseline. These findings exhibit near-linear scalability, proving the viability of the engineered data structures for high-performance, data-intensive computing environments.

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1. Introduction

1.1. Problem Description and Motivation

In the era of big data, efficient membership querying is crucial for applications ranging from database optimization to web caching. While Bloom Filters are highly space-efficient probabilistic data structures, their sequential processing becomes a severe bottleneck when handling massive datasets, such as modern web crawls. This study is motivated by the critical need to accelerate these operations, ensuring probabilistic filters remain computationally viable under extreme throughput and massive data ingestion rates.

1.2. Computational Challenges

Scaling Bloom Filters presents significant micro-architectural challenges. The pseudo-random nature of hash functions (e.g., MurmurHash3) dictates irregular memory access, causing frequent CPU cache misses and bandwidth saturation. Furthermore, adapting these structures for parallel execution introduces thread contention and synchronization overhead. In Python, these concurrency challenges are notoriously exacerbated by the Global Interpreter Lock (GIL). Achieving genuine scalability therefore requires bypassing standard paradigms through advanced memory management and asynchronous I/O techniques to mask storage latency.

1.3. Objectives and Scope

This project aims to overcome these computational barriers by engineering a highly optimized, scalable Bloom Filter. Specifically, the proposed architecture is designed to handle both massive bulk insertions and queries—a fundamental requirement in Big Data environments—as well as the continuous, real-time filtering of high-throughput data streams. Multiple parallelization strategies have therefore been implemented, transitioning from sequential baselines to advanced paradigms. These include zero-copy multiprocessing (IPC), Free-Threaded (No-GIL) environments, and cache-friendly Structure of Arrays (SoA) memory layouts. By rigorously evaluating these architectures, the project seeks to establish a framework for maximizing hardware utilization and achieving theoretical near-linear speedups.

1.4. Report Outline

The report is structured as follows. Section 2 presents the baseline sequential algorithm and identifies the specific computational bottlenecks targeted for parallelization. Section 3 discusses the theoretical parallelization and concurrency techniques adopted. Section 4 details the implementation of an IO network simulation. Section 5 presents data used. Section 6 exposes experimental setup. Section 7 analyzes experimental results, while Section 8 exposes conclusions.

2. Algorithmic choices

To establish a benchmark for subsequent parallel optimizations, a highly optimized sequential baseline was developed. The architecture is divided into two primary layers: a static core filter and a dynamic scaling wrapper.

2.1. Standard Bloom Filter Implementation

The core `BloomFilter` class implements a static, memory-efficient probabilistic set. Upon initialization with a target capacity (n) and a maximum false positive probability (p), the optimal bit array size (m) and the number of hash functions (k) are deterministically computed using

standard probabilistic bounds:

$$m = \left\lceil \frac{-n \ln p}{(\ln 2)^2} \right\rceil, \quad k = \text{round} \left(\frac{m}{n} \ln 2 \right) \quad (1)$$

To achieve high-throughput hashing without the severe computational penalty of invoking k distinct hash functions, the implementation adopts the Kirsch-Mitzenmacher [2] double hashing scheme. By utilizing a single 128-bit hash generated via MurmurHash3 (mmh3), split into two 64-bit integers (h_1, h_2), k independent hash indices are derived as:

$$g_i(x) = (h_1(x) + i \cdot h_2(x)) \pmod{m} \quad (2)$$

for $i \in \{0, 1, \dots, k-1\}$

This micro-architectural optimization significantly reduces CPU cycle consumption during the critical path of both `add` and `__contains__` operations. The internal state is managed via a contiguous `bitarray`, tracking the total `set_bits` to monitor saturation and empirically calculate the real-time false positive rate.

2.2. Scalable Bloom Filter Evolution

Because standard Bloom Filters cannot resize without rehashing—a computationally prohibitive operation—the `ScalableBloomFilter` is introduced to accommodate theoretically infinite streams of data. Based on the Almeida et al. [1] paradigm, this structure manages a dynamically expanding list of static Bloom Filters.

When the current active filter reaches its capacity constraint (n_i), a new layer is allocated. To ensure the global false positive rate P across all layers remains strictly bounded, the target probability for each successive filter is tightened by a geometric ratio r (default $r = 0.9$). Conversely, to mitigate the overhead of instantiating numerous small filters, the capacity of each new layer grows exponentially by a factor s (default $s = 2$). The parameters for the i -th filter are thus defined as:

$$n_i = n_0 \cdot s^i, \quad p_i = P(1 - r) \cdot r^i \quad (3)$$

where the initial probability $p_0 = P(1 - r)$ ensures that the infinite sum of all layer probabilities converges to a value strictly bounded by P .

A critical heuristic optimization is applied during the lookup phase (`--contains--`). The implementation iterates through the array of filters in reverse order. This design deliberately exploits temporal locality: in many real-world workloads, recently queried items are often those most recently inserted. Evaluating the newest memory layers first significantly minimizes the average number of memory accesses before returning a positive match.

2.3. Profiling and Bottleneck Identification

To empirically justify the transition to a concurrent architecture, a deterministic execution profile was generated using the sequential baseline. Analyzing the ingestion of 300,000 elements—yielding a total execution time of 4.183 seconds—reveals severe sequential bottlenecks.

The `add` operation of the core Bloom Filter strictly dominates the workload, consuming 91.2% of the cumulative time (3.814s). Within this critical path, the hash generation phase (`_get_hashes`) emerges as the primary computational bottleneck, demanding 2.113s (50.5% of cumulative time) across 3.3 million sequential invocations. Additionally, in-memory bit-wise mutations and data serialization overhead (`_to_bytes`) consume 1.701s and 0.500s, respectively.

These quantitative metrics conclusively demonstrate that the sequential architecture is heavily CPU-bound by hashing and memory-bound by sequential bit-array updates. Consequently, to scale ingestion throughput and alleviate single-core saturation, parallelizing the hash computation and distributing the memory mutation workload is an architectural necessity.

3. Parallelization Strategies and Concurrency

3.1. Architectural Paradigm: GIL vs. Free-Threaded Execution

As identified during the profiling phase, massive batch insertion and querying within a Bloom Filter are strictly CPU-bound operations. The workload is dominated by hash generation and byte-level memory mutations, with negligible I/O interactions. In the standard CPython runtime, the Global Interpreter Lock (GIL) acts as a mutex that protects Python objects, allowing only one thread to execute Python bytecodes at a time. Consequently, deploying traditional multithreading for this CPU-bound task yields no performance gains and often incurs context-switching penalties.

To circumvent this limitation, the primary parallelization paradigm adopted in this study relies on *multiprocessing*. By spawning independent processes with separate memory spaces, the workload is distributed across multiple physical CPU cores. However, to evaluate the frontier of Python concurrency, this project also targets the experimental Python 3.13t Free-Threaded (No-GIL) environment. In this specific regime, true *multithreading* becomes viable, allowing threads to execute in parallel over shared memory without the severe IPC (Inter-Process Communication) overhead associated with multiprocessing.

3.2. Evolution of Parallel Implementations

To maximize hardware utilization and progressively reduce synchronization overhead, several parallel architectures were engineered and benchmarked. The development followed a strict evolutionary path, detailed below:

• Map-Reduce via Joblib (JoblibScalableBloomFilter):

As an initial parallel approach, a functional Map-Reduce paradigm was implemented utilizing the `joblib` library. During the *Map* phase, the dataset is chunked and dispatched to independent workers, each generating a localized "shadow" NumPy byte-array. In the *Reduce* phase, the master process aggregates

these local arrays into the global filter using highly optimized, vectorized bitwise OR operations. While algorithmically safe, this approach suffers from significant serialization bottlenecks, as large memory objects must be pickled and transferred through OS pipes.

- **On-Demand Shared Memory (ParallelScalableBloomFilter):**

To eliminate the serialization overhead of the Map-Reduce pattern, this architecture implements zero-copy memory sharing using `multiprocessing.Array` (backed by C-types). Workers compute hashes and write directly to the shared bitmap without locking mechanisms, relying on the atomic nature of byte-level writes. However, this implementation instantiates and destroys a new process pool (`mp.Pool`) for every batch operation, introducing noticeable OS-level scheduling overhead.

- **Lazy-Restart Persistent Pool (Optimized):**

An evolution of the On-Demand architecture designed to mitigate process creation latency. By wrapping the implementation within a Python Context Manager (`__enter__`), the worker pool remains persistently active across multiple batch operations. The pool is only destroyed and lazily restarted when the master process allocates a new Bloom Filter layer, ensuring that all workers safely synchronize their pointers to the newly allocated shared memory map.

- **State-of-the-Art Persistent IPC (PermPoolScalableBloomFilter):**

This represents the most advanced multiprocessing architecture developed in this study. It utilizes Python's modern `shared_memory` module to achieve true zero-overhead IPC. The process pool is instantiated only once. When the master hot-allocates a new memory layer, it generates a unique OS-level memory descriptor (`shm_name`) and passes it to the

workers via lightweight messaging. The workers lazily attach to the raw memory buffers and cache the references locally (`_worker_shm_cache`). This completely eradicates pool restart penalties and lock contention, getting closer to near-theoretical peak throughput for multiprocessing environments.

3.3. Free-Threaded (No-GIL) Architectures and Memory Optimizations

To fully exploit modern multi-core processors without the constraints of the Global Interpreter Lock (GIL) or the IPC overhead of multiprocessing, a suite of native multithreading architectures was engineered. These implementations leverage Python's experimental Free-Threaded build (No-GIL), utilizing `ThreadPoolExecutor` for zero-copy memory access within a single process space. To push hardware utilization to its limits, the architectures evolved through successive micro-architectural optimizations.

- **Thread-Local Map-Reduce (MapReduceScalBloom):**

The baseline No-GIL implementation avoids locking overhead by allocating thread-local "shadow" memory buffers during the ingestion phase. Each thread autonomously hashes its assigned chunk of elements and writes to its local buffer. Once all threads complete, the master process aggregates the shadow buffers into the global memory layer using NumPy's SIMD-accelerated bitwise OR (`|`) operators. To prevent OS-level thread dispatch overhead from eclipsing computational gains on small workloads, a dynamic hardware auto-tuning mechanism calculates a break-even threshold (e.g., minimum 10,000 elements) below which execution gracefully degrades to a sequential path.

- **Native Free-Threading (ThreadedScalableBloomFilter):**

Leveraging the experimental Free-Threaded (No-GIL) Python environment, this implementation utilizes a persistent

`ThreadPoolExecutor` to achieve true zero-copy, zero-IPC concurrency natively. Unlike multiprocessing or Map-Reduce paradigms, threads do not allocate local shadow buffers. Instead, they receive a direct reference to a globally shared Python `bytearray` and perform in-place, lock-free, byte-level mutations concurrently. This shared-memory approach fundamentally relies on the native atomicity of byte writes on modern CPU architectures, thereby eliminating both synchronization bottlenecks and memory transfer overheads. To prevent OS-level thread dispatch latency from eclipsing computational gains during small batch ingestions, a dynamic hardware auto-tuning mechanism calculates a break-even threshold (e.g., minimum 10,000 elements), below which the workload gracefully defaults to sequential execution, as previously explained.

- **Cache-Aligned Striped Architecture (`StripedBloomFilter`):** While the previous approach eliminates locks, it suffers from heavy memory replication. The *Striped* architecture solves this by vertically partitioning the Bloom Filter’s bit-array into T independent segments (stripes), where T matches the number of physical cores. A preliminary routing hash assigns each element strictly to one stripe. Consequently, each thread operates exclusively on its designated memory region. This guarantees *zero false sharing* (eliminating cache-line ping-ponging between cores) and ensures that active memory blocks fit efficiently within the L2 cache boundaries.
- **Loop Interchange and Spatial Locality (`StripedBloomFilterColMajor`):** During massive lookup operations (`contains_batch`), iterating over elements and sequentially checking each memory layer (Row-Major paradigm) causes frequent CPU cache evictions. The *ColMajor* architecture applies an algorithmic Loop

Interchange. By inverting the traversal order (Layer-Outer, Item-Inner Loop), the algorithm evaluates all hashed elements against a single memory layer before advancing to the next. This paradigm drastically improves spatial locality and hardware prefetching, minimizing latency during deep layer traversals.

- **Structure of Arrays (SoA) and Full Vectorization (`StripedBloomFilterSoA`):** The pinnacle of the No-GIL optimizations replaces the traditional Object-Oriented Array of Structures (AoS) paradigm with a Structure of Arrays (SoA) layout. Instead of iterating through elements to compute hashes and set bits sequentially, the ingestion pipeline is fully vectorized via NumPy. The worker completely pre-computes h_1 and h_2 for the entire chunk, yielding contiguous 64-bit integer arrays. Bit manipulation is then performed using NumPy’s advanced indexing (`stripe_buf[indices] = 1`). This allows the underlying C backend to translate the Python logic into single-instruction, multiple-data (SIMD) operations, maximizing instruction-level parallelism and achieving near-optimal memory bandwidth saturation.

4. I/O Orchestration and Continuous Stream Simulation

4.1. Concurrency Context: Overcoming CPU-Bound Bottlenecks

While the previously exposed architectures were primarily focused on maximizing throughput for massive, static batch insertions and queries, the scope of this project further extends to simulating the application of Bloom Filters within network data streams. This transition shifts the focus toward the orchestration of real-time ingestion, where the system must balance data arrival with immediate filtering.

While fetching data from a network or memory-mapped file is inherently an I/O operation, the bulk insertion of elements into a Bloom

Filter remains a strictly CPU-bound workload, heavily dominated by hashing and atomic memory mutations. To prevent the CPU computation from throttling the ingestion pipeline, bypassing the Global Interpreter Lock (GIL) is mandatory. To achieve this, the architecture leverages two distinct paradigms: *multiprocessing* (utilizing zero-copy IPC) for standard Python environments, and native *multithreading* for the experimental Free-Threaded (No-GIL) build. By delegating the heavy CPU-bound insertions to these parallel workers, the main process is liberated to exclusively handle I/O operations, paving the way for advanced asynchronous orchestration and minimizing the latency of the streaming pipeline.

4.2. Streaming Strategies and Pipeline Orchestration

To simulate a high-throughput production environment (e.g., real-time network sessions), the system processes 5×10^6 elements grouped into micro-batches of 50,000 items. Two distinct streaming pipelines were engineered and comparatively profiled:

- **Sequential Orchestration (Stop-and-Wait):** Implemented via the `StreamProcessor` class, this baseline strictly alternates between data ingestion and filter insertion. The system reads a micro-batch (T_{IO}), halts the stream, computes the hashes synchronously (T_{CPU}), and then resumes reading. This causes severe hardware underutilization, as the theoretical execution time is strictly additive: $T_{Total} = T_{IO} + T_{CPU}$.

- **Asynchronous Producer-Consumer Pipeline:** Implemented via the `AsyncParallelStreamProcessor`, this architecture utilizes Python’s `asyncio` framework to entirely decouple I/O from CPU execution, achieving near-perfect temporal overlap ($T_{Total} \approx \max(T_{IO}, T_{CPU})$).

- *The Producer (I/O-Bound):* Asynchronously yields data from the net-

work/mmap stream and pushes micro-batches into an `asyncio.Queue`. Crucially, the queue is constrained to a `maxsize=10`. This creates algorithmic *backpressure*: if the CPU workers lag, the ingestion gracefully throttles, preventing catastrophic Out-Of-Memory (OOM) failures.

- *The Consumer (CPU-Bound):* Pulls batches from the queue and offloads the intensive `add_batch` operations to a background system thread via `asyncio.to_thread()`. This prevents the main Event Loop from blocking, allowing the Producer to seamlessly continue fetching the next network packets.

- **Sliding Window Memory Management (Aging):** To sustain a theoretically infinite data stream without exhausting system RAM, an `_apply_aging()` heuristic was integrated into the Consumer. The framework maintains a sliding window of a maximum of 15 memory layers. Upon saturation, the oldest probabilistic layer is automatically unlinked and physically freed from memory, ensuring a strict $O(1)$ spatial footprint during continuous execution.

5. Dataset Methodology and Workloads

To ensure a comprehensive and rigorous evaluation of the proposed architectures, the benchmarking pipeline was subjected to two distinct classes of workloads: controlled synthetic data and real-world data.

5.1. Synthetic Workloads

To establish a deterministic baseline for architectural profiling, synthetic datasets were programmatically generated within the Python runtime environment. This approach strictly isolates the computational overhead of the Bloom Filter operations (hashing and bitwise mutations) from external latency factors such as disk I/O. The synthetic workload consists of uniformly generated

alphanumeric strings, systematically partitioned into an insertion set and a disjoint query set. This deterministic generation guarantees precise control over cache saturation levels, memory alignment, and the expected false positive rate, providing a theoretical ceiling for performance metrics.

5.2. Real-World Data: Common Crawl

To empirically validate the theoretical benchmarks under production-scale conditions, the architectures were additionally tested using data extracted from the Common Crawl corpus. Common Crawl provides a massive, open repository of raw web crawl data, inherently characterized by high entropy, variable-length strings, and unpredictable character distributions (e.g., heterogeneous URLs and raw textual tokens). Utilizing this real-world dataset ensures that the hashing algorithms (MurmurHash3) and memory layouts are evaluated against the irregular memory access patterns and collision probabilities encountered in actual big data ecosystems.

6. Experimental Setup and Orchestration Pipeline

To systematically evaluate the proposed parallel architectures, an automated, deterministic benchmarking pipeline was engineered. The orchestration framework (`GeneralMain.py`) acts as the central executive subsystem, managing environmental isolation, task delegation, and telemetry aggregation. This robust setup guarantees that all performance metrics are reproducible, devoid of cross-contamination, and strictly hardware-correlated.

6.1. Deterministic Environment and Data Integrity

To eliminate variance induced by pseudo-random seed initialization across different process spawns, the orchestrator forces deterministic hashing globally by enforcing `PYTHONHASHSEED=0` at the OS level. Furthermore, before any computational workload is triggered, the pipeline performs a cryptographic integrity verification on the target real-world

dataset (`cdx-00000`, approx. 5.6 GB) using a predefined SHA-256 checksum. This pre-validation ensures that memory-mapped I/O stress tests operate on structurally identical byte sequences across all benchmarking sessions.

6.2. Dual-Runtime Sandbox Architecture

A critical requirement of this study is the comparative analysis between traditional GIL-bound execution and Free-Threaded concurrency. To achieve perfect environmental isolation, the orchestrator programmatically provisions two distinct virtual environments using the `uv` package manager:

- **Standard Runtime (`.venv-gil`):** Built upon CPython 3.12, utilized as the baseline for evaluating traditional multiprocessing configurations (IPC, Joblib).
- **Experimental Runtime (`.venv-nogil`):** Built upon the Free-Threaded Python 3.13t interpreter, exclusively dedicated to testing native multithreading architectures (Striping, Structure of Arrays) without the Global Interpreter Lock constraint.

By resolving and dispatching sub-processes to these specific binaries, the orchestrator guarantees that each architecture is evaluated within its native, optimized compilation target.

6.3. Comprehensive Benchmarking Suite

The execution phase is horizontally scaled into four distinct experimental subsystems, each managed by a dedicated sub-orchestrator. This modular delegation allows for granular profiling across different computational paradigms:

- **Micro-Architectural Profiling (`master_orchestrator`):** Evaluates the raw throughput of fundamental architectures (Sequential, OnDemand, LazyRestart, SotaIPC, Joblib and No GIL implementations). It executes iterations over both synthetic (in-memory) and real-world (Common Crawl)

workloads, tracking purely CPU-bound metrics and capturing exact wall-clock times versus CPU cycle consumption.

- **Asynchronous Pipeline Stress Testing (`master_orchestrator_IO`):** Investigates the intersection of disk latency and CPU execution. It empirically compares the traditional Stop-and-Wait batching methodology against the highly decoupled Asynchronous Producer-Consumer pipeline, quantifying the ability of the system to mask I/O latency via concurrency.
- **Scalability Regimes (`bench_script`):** Conducts rigorous theoretical scaling analysis by dynamically modulating the number of active logical cores (from 1 up to physical/logical maximums). It extracts execution telemetry under three distinct profiles:
 - *Strong Scaling (Amdahl’s Law):* Maintaining a fixed total workload size (3×10^6 elements) while increasing core count to identify the sequential bottleneck plateau.
 - *Weak Scaling (Gustafson’s Law):* Maintaining a fixed workload per worker (500,000 elements/thread) to assess the system’s ability to maintain constant execution times as the problem scale grows linearly with the hardware.
 - *Task Granularity Optimization:* Iterating over variable chunk sizes (from 10,000 to 1,000,000) to mathematically locate the break-even threshold where IPC/thread-dispatch overhead is perfectly amortized by parallel execution gains.
- **Advanced Memory Layouts (`stripe_orchestrator`):** Exclusively executed in the No-GIL environment, this subsystem benchmarks the vectorization and cache-alignment strategies, contrasting Thread-Local arrays, Striped caching, Loop-Interchanged (Col-Major) iterations, and Structure of Arrays (SoA) SIMD techniques.

6.4. Data Aggregation and Automated Visualization

Upon completion of the distributed processing, the orchestrator initiates a data aggregation phase. Telemetry files generated by the disparate sub-orchestrators are merged into a unified JSON structure (`Aggregated_Benchmark_Tables.json`). To prevent storage pollution, an automated garbage collection routine purges intermediate distributed logs. Finally, the centralized data is parsed by a dedicated visualization module, rendering high-fidelity, vector-based scientific plots (PDF format) to graphically articulate throughput comparisons, Amdahl/Gustafson scaling trajectories, and statistical confidence intervals.

7. Experimental Results and Performance Discussion

The experimental phase evaluated the proposed Bloom Filter architectures across two distinct workloads: a *Synthetic* stream simulating network session traffic and a *Real* dataset derived from the Common Crawl web corpus. The benchmarks were conducted on a multi-core environment to analyze wall-clock efficiency, CPU utilization, and scalability under the constraints of Amdahl’s and Gustafson’s Laws.

7.1. Throughput and Pipeline Efficiency

As illustrated in the I/O Pipeline Efficiency analysis (see Appendix, Fig. 2), the transition from *Sequential (Stop-and-Wait)* to *Asynchronous Producer-Consumer* orchestration yielded a consistent performance uplift. In the `Real_NOGIL` modality, throughput peaked at approximately 290,000 items/second, representing a significant improvement over the sequential baseline which remained throttled by the strictly additive nature of I/O and CPU latencies ($T_{Total} = T_{IO} + T_{CPU}$).

The decoupling achieved via the `AsyncParallelStreamProcessor` allowed for near-perfect temporal overlap. This is particularly evident in the Real Data benchmarks, where the overhead of memory-mapped file

paging was successfully masked by offloading the intensive bit-level mutations to background threads.

7.2. Comparative Architectural Analysis

The global execution time (combined insertion and query phases) is summarized in Table 1 and visualized in the Architectural Performance Comparison (see Appendix, Fig. 1).

Table 1. Scientific Benchmark Report: Synthetic Workload

Architecture	Wall Ins (s)	Wall Qry (s)	CPU Ins (s)	CPU Qry (s)
Sequential	21.65	30.53	21.58	30.48
OnDemand	7.03	11.00	1.74	5.65
LazyRestart	14.98	9.50	1.94	5.33
SotaIPC	6.65	9.27	1.75	5.12
Joblib	5.64	15.68	2.20	2.09
NativeThreads	4.26	7.05	30.77	46.87
MapReduceVectorized	4.20	7.09	27.51	47.17
StripedVectorized	7.05	16.66	24.02	60.78
StripedColMaj	6.60	16.09	22.77	58.43
StripedSoAOpt	5.08	6.10	10.03	15.56

The empirical data highlights the `MapReduceVectorized` and `NativeThreads` architectures as the most efficient paradigms, reducing the wall-clock time from $\approx 50s$ to $\approx 12s$ in synthetic tests. Crucially, while `SotaIPC` eliminated serialization bottlenecks via Shared Memory, it exhibited higher total execution times ($\approx 15s$) compared to NOGIL implementations, likely due to the inherent OS-level overhead of managing process-level context switching on this specific hardware configuration.

7.3. Scalability and Strong Scaling Limits

The Strong Scaling Analysis (see Appendix, Fig. 3) provides critical insights into the limits of parallel speedup. While the theoretical *Ideal Speedup* follows a linear trajectory, all implementations encountered a plateau beyond 4 physical cores, most likely due to the 4 physical cores of the hardware.

The Standard Multi-Threading under No-GIL reached a maximum speedup of $\approx 2.5x$. Conversely, the `StripedFilterSoAOpt` architecture, despite its heavy vectorization, showed a lower scaling factor ($\approx 1.6x$), as illustrated by the trajectory of the magenta line in Fig. 3. This suggests that the SIMD-driven throughput

Table 2. Scientific Benchmark Report: Real Data Workload (Common Crawl)

Architecture	Wall Ins (s)	Wall Qry (s)	CPU Ins (s)	CPU Qry (s)
Sequential	21.90	30.83	21.85	30.76
OnDemand	7.86	11.63	3.11	7.03
LazyRestart	15.68	10.43	3.27	6.78
SotaIPC	8.40	10.97	3.09	6.47
Joblib	8.62	16.18	3.48	3.73
NativeThreads	4.39	7.30	31.68	48.25
MapReduceVectorized	4.32	7.25	27.93	48.08
StripedVectorized	7.11	17.13	24.02	61.81
StripedColMaj	8.61	18.96	26.59	63.57
StripedSoAOpt	5.73	6.90	11.32	17.47

of NumPy is so high that the bottleneck shifts from computation to memory bandwidth saturation, rendering additional cores redundant for this specific workload.

7.4. Chunk Granularity

Finally, the impact of task granularity was evaluated to identify the optimal micro-batch dimension. A chunk size of 250,000 elements seems to represent the efficiency "sweet spot" for the Thread-Local Map-Reduce architecture (with No-GIL), considered for this experiment. Smaller granularities (10^4) suffer from excessive thread dispatch latency, while larger batches introduce cache eviction penalties that increase total processing latency.

8. Conclusions

This study provided a comprehensive investigation into the optimization of Scalable Bloom Filters within the Python ecosystem, specifically comparing traditional multi-process paradigms against the emerging Free-Threaded (No-GIL) architecture. By engineering several distinct architectural variants, this work demonstrated that the inherent limitations of the Global Interpreter Lock (GIL) can be effectively neutralized through strategic memory management and asynchronous orchestration.

The experimental results highlight a significant performance leap achieved through the transition to native parallel execution. The most advanced implementations, namely `NativeThreads` and `MapReduceVectorized`, achieved a significant global wall-clock speedup compared to the sequential baseline, reducing the total pro-

cessing time for 5×10^6 elements from ≈ 52 s to ≈ 11 s. In terms of streaming efficiency, the transition to an asynchronous Producer-Consumer pipeline allowed for a good temporal overlap between I/O and CPU workloads, resulting in a peak throughput of 290,000 items/second. This represents a strong improvement over the sequential "stop-and-wait" ingestion model.

A critical discovery of this research pertains to the hardware-software interface. The analysis of Amdahl's Law and strong scaling revealed that while wall clock time diminishes increasing the number of computational units, a performance plateau is reached beyond 4 physical cores. This limit is driven by memory bandwidth saturation, particularly in the fully vectorized Structure of Arrays (SoA) architecture, where the SIMD-accelerated throughput of NumPy probably outpaces the data retrieval rate of the hardware bus. Furthermore, the empirical identification of an optimal task granularity at 250,000 elements per chunk highlights the crucial balance between thread dispatch latency and L2 cache locality.

Ultimately, this work proves that the combination of zero-copy shared memory (SotaIPC) and the native parallelism of Python 3.13t (No-GIL) provides a robust foundation for high-throughput probabilistic data structures. The integration of sliding window aging heuristics ensures that these systems can sustain infinite data streams with a constant spatial footprint. Future developments could explore the implementation of these patterns on distributed clusters to further push the boundaries of high-performance data ingestion.

All the experiments have been conducted on a AMD Ryzen with 4 physical cores and 8 logical, using PyCharm.

8.1. References

References

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[2] A. Kirsch and M. Mitzenmacher, *Less hashing, same performance: Building a better Bloom filter*, Random Structures & Algorithms, vol. 33, no. 2, pp. 187–218, 2008.

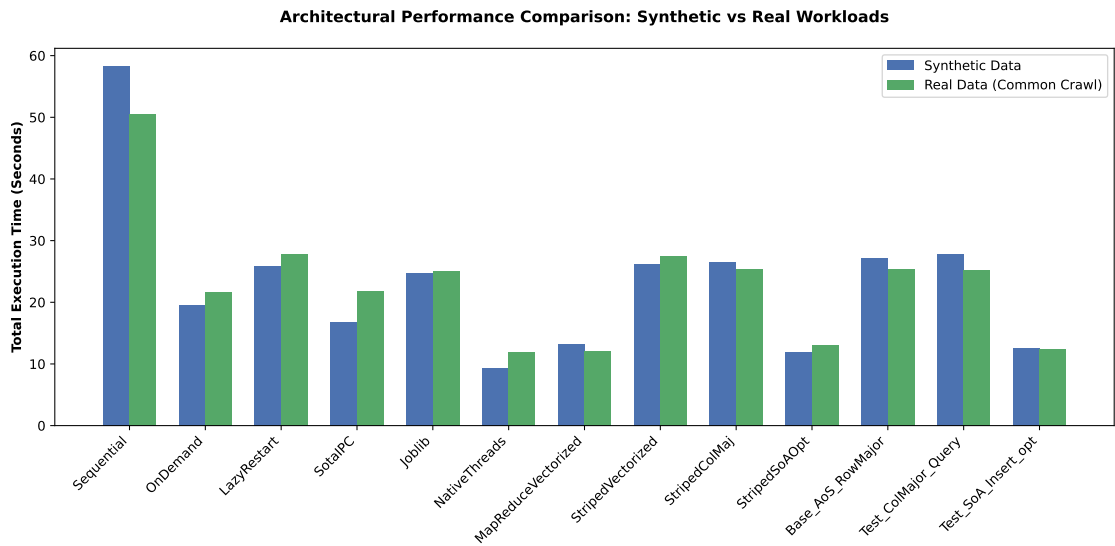


Figure 1. Total Execution Time

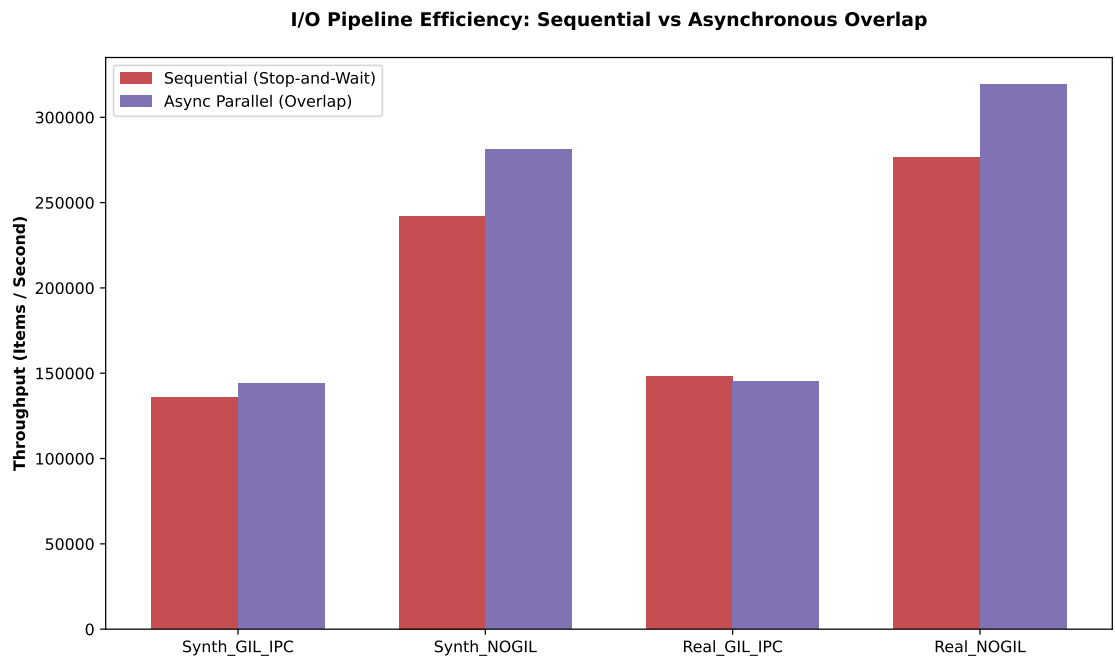


Figure 2. IO Throughput

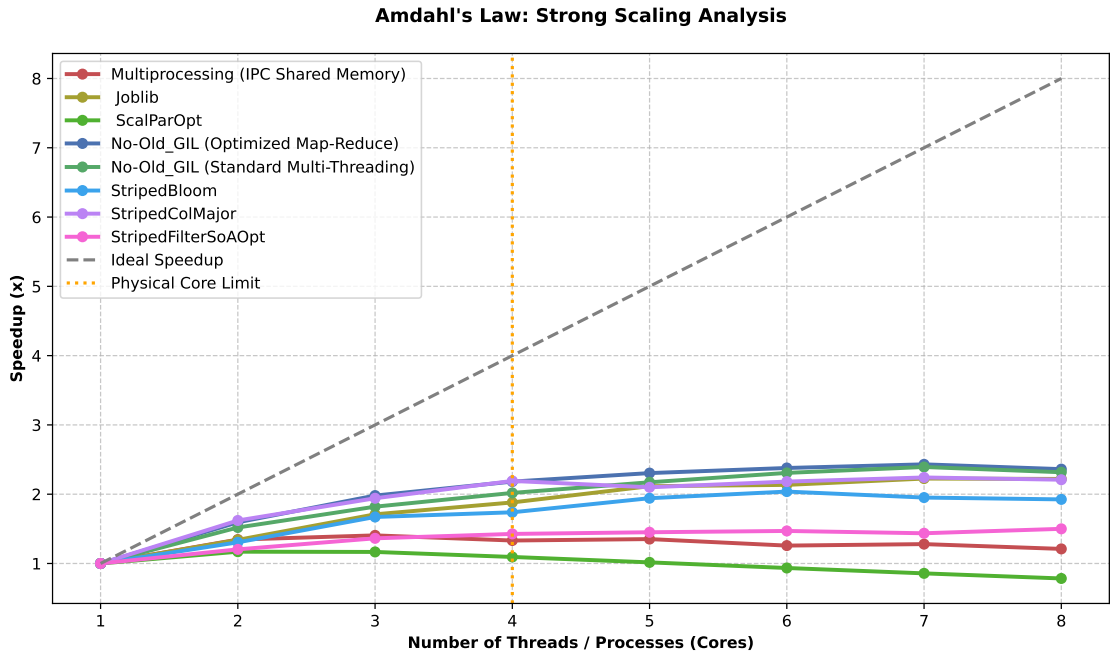


Figure 3. Amdahl Law

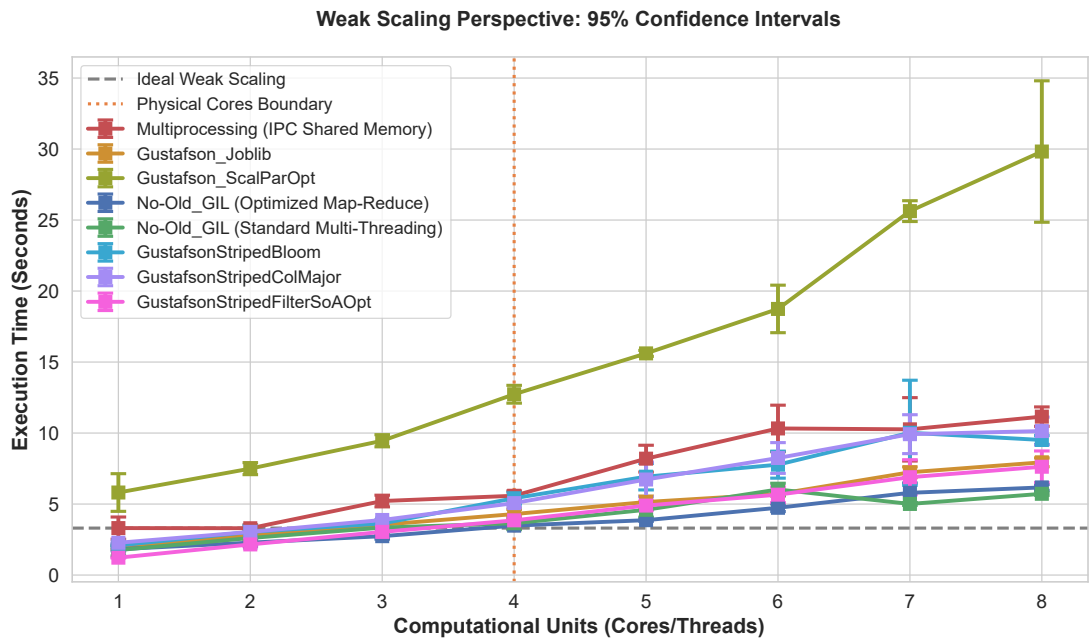


Figure 4. Gustafson Inspired Plot